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COGNITIVE FALLACIES

We have started this series of cognitive fallacies in the class. Most of the methods to understand this series is going to be in class but for your notes, I have only included from exam point of view.

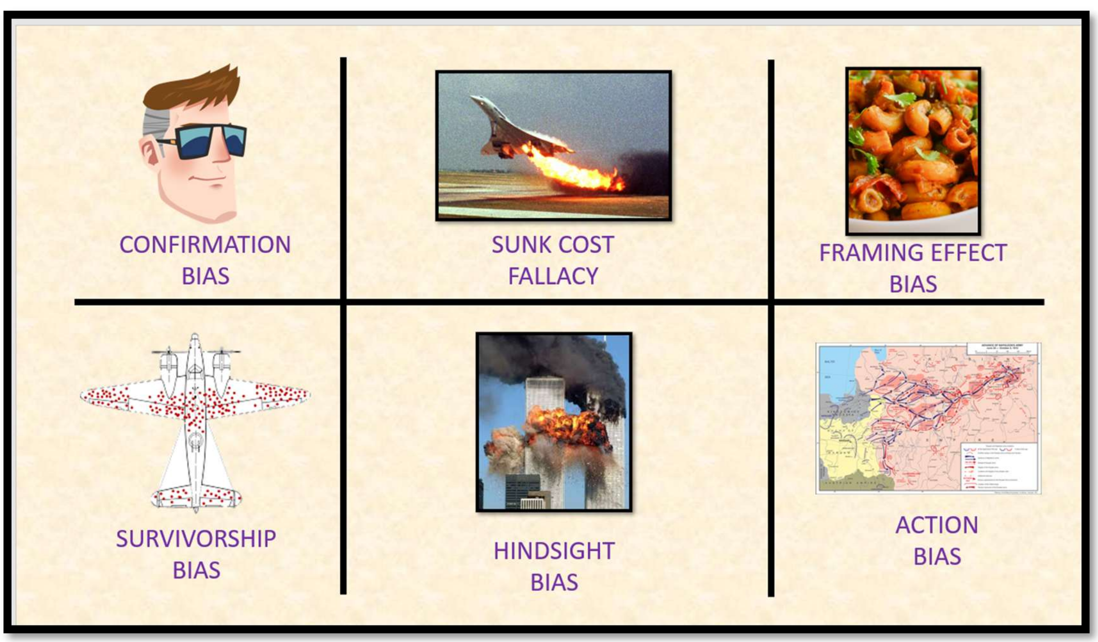
In the beginning of the class, we started with the experiment of bidding (Auction) where we discussed how to become a smart person. This smartness is not to deceive others but to not do similar mistakes in our life. To understand this, we need to understand the broader sense of **cognitive fallacy.**

So, what is Cognitive fallacy?

Cognitive fallacy is **a systematic pattern of ERRORS in Thinking and Decision Making.** It is related to Memory, Attention, Attribution, and other mental mistakes. A cognitive bias is an inherent thinking "blind spot" that can lead to inaccurate and often irrational conclusions. These biases affect belief formation, reasoning processes, business decisions, and human behaviour. Unlike logical fallacies, which stem from errors in logical arguments, cognitive biases arise from thought processing errors—often related to memory, attention, attribution, and other mental factors.

We will at least go through 10+ cognitive fallacies.

We went through and discussed 6 cognitive fallacies which are:



I. Confirmation Bias:

- This cognitive bias affects decision-making, research, and everyday interactions, leading us to maintain or strengthen our beliefs even in the face of contrary evidence.
- Tendency to favor information that confirms our beliefs.
- Interpret, remember evidences which we already started believing in.
- It is like wearing glasses.

Let's see 9 Examples to understand deeper.

1. **The condition when someone doesn't like you:**

- Imagine you're convinced that a coworker dislike you. When they don't greet you in the morning, you interpret it as evidence supporting your belief. However, you ignore the times they've been friendly or busy with their own thoughts.
- *Lunch Breaks,*
- *Meeting reactions,*
- *Messages or mail responses,*
- *Body language,*
- *Gatherings,*
- *Social media interactions*

2. **Politics and News Consumption:**

- Suppose you follow a particular news channel that aligns with your political views. When they report on a controversial topic, you accept their perspective without critically evaluating other sources that might offer a different viewpoint.
- Arnab Goswami, Dhruv Rathee etc.
- Modi or Rahul Gandhi
- BJP or Congress

3. **Choosing Friends and Social Circles:**

- You join a hobby group where everyone shares your passion. Over time, you bond with like-minded individuals, reinforcing your existing beliefs. You may even dismiss dissenting opinions within the group.
- **Bonding with like-minded people and nudging out other ideas from other people.**
- **Coming from same economic class / geography / language**
- **Dismissal of Dissent**

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4. **Selective Memory:**

- After a heated argument, you remember every detail that supports your stance. The opposing arguments fade away, leaving you with a vivid memory of your own points.
- **Online Product Reviews**
- You read reviews online, seeking validation for your initial preference.
- Confirmation bias kicks in when you selectively focus on positive reviews that align with your chosen brand or model.
- You might ignore negative reviews or attribute them to individual experiences or biases.
- As a result, your decision becomes influenced by the information that confirms your preconceived notion about the smartphone's quality.

5. **Investment Decisions:**

- You're convinced that a particular stock will soar. You eagerly seek out positive news about the company, ignoring any negative indicators. This bias can lead to risky investment choices.
- **Adani shares and Hidenburg research report**
- **Elections and Stock market**
- **Ignoring Contradictory Signals**
- **Seeking Confirmation but positive to your investments.**

6. **Health and Wellness:**

- If you believe a specific diet is effective, you'll notice stories of people who lost weight following it. You might overlook cases where the diet didn't work or caused health issues.
- **Fad diets – trendy diet plans**
- **Supplements and miracle pills – advertisements**
- **Exercise routines – HIIT and you start searching similar reels.**
- **Detox diets – ignoring warnings**

7. **Relationships and Compatibility:**

- When dating, you focus on your partner's positive traits, attributing their flaws to external factors. You downplay any evidence that challenges your perception of compatibility.
- **Idealization of new partner**
- **Ignoring 'Red Flags'**
- **Selective memory in Relationships**
- **Comparing to past relationships**

8. **Job Interviews:**

- As an interviewer, you form an initial impression of a candidate. Throughout the interview, you subconsciously seek evidence that confirms your impression, potentially overlooking valuable skills.
- **Subconscious focus on finding faults in the first impression bias.**
- **Focusing on nervousness or hesitations**
- **Wrong assessment**

9. **Scientific Research Interpretation:**

- Scientists analyzing data related to climate change may emphasize findings that align with their hypotheses. They might downplay contradictory data, affecting the overall research conclusions.
- **Climate change research – downplay or dismiss other variables**
- **Drug Efficacy studies - Positive results (e.g., symptom improvement) are highlighted and Negative outcomes (side effects, lack of efficacy) might be minimized.**
- **Biased samplings in Surveys**
- **Publication bias in journals.**

The Way Forward:

- BE AWARE
- CONSIDER ALL EVIDENCES
- STAY OPEN-MINDED
- LOOK FOR OPPOSITE INFORMATION
- INTERACT WITH DIFFERENT TYPE OF PEOPLE
- नाि◌Öत िवा◌ा समं चe◌ु ।
- Knowledge is Power.
- आ नो भp◌ा◌ः . . .

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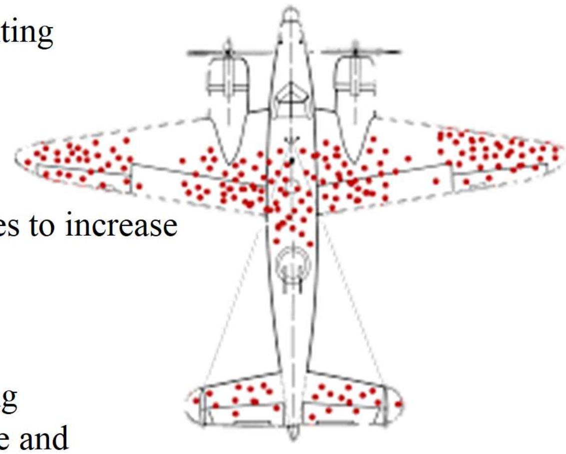
2. Survivorship Bias

Let's go through the dilemma that came before aircraft engineers in II world war.

The Context:



During World War II, military planners faced the daunting task of improving the survivability of their aircraft. They needed to analyse the damage patterns on planes that returned from combat missions.

The goal was to determine where to reinforce the planes to increase their chances of making it back safely.

The Data Collection:

Military engineers meticulously examined the surviving planes—the ones that managed to withstand enemy fire and return to base.

These surviving planes displayed damage primarily in areas that were less critical, such as wings and tail sections.

The Flawed Assumption:

The planners made a critical assumption: if the planes survived, it meant that the damage they sustained was manageable. Therefore, they recommended reinforcing those specific areas.

After all, it seemed logical to focus on strengthening the parts that had successfully endured combat.

The Missing Data:

However, there was a glaring oversight—the planes that didn't return. These were the ones that were shot down during combat.

These lost planes had critical damage in different areas, including engines, cockpit, and fuselage. By ignoring this missing data, the planners inadvertently reinforced the wrong parts.

The Correct Approach:

To address survivorship bias, military experts needed to consider both the surviving and crashed planes. Analysing the entire dataset was crucial.

By doing so, they would have realized that critical areas needed reinforcement—areas that were vulnerable and led to the planes' demise.

Reinforcing these vulnerable spots would have significantly improved overall aircraft survivability.

The Lesson:

Survivorship bias teaches us to look beyond the survivors. Success stories alone don't provide the complete picture.

Whether in military strategy, investment analysis, or any other field, we must account for the entire spectrum of outcomes, including failures.

- Paying attention to only those who have succeeded, ignoring the thousands that failed.
- By focusing on survivor planes in WWII, rather than the failed planes.
- Overestimating historical performances when analyzing things.

Let's understand this with more examples:

Example 1:

- **Business / Restaurant success-review stories.**

When we read glowing reviews about successful businesses or restaurants, we often forget the ones that didn't make it.

Survivorship bias leads us to overvalue the positive stories while ignoring the lessons from failures.

For every thriving restaurant or business, there are many that struggled, closed down, or faced challenges.

Takeaway: To make informed decisions, consider both the success stories and the untold narratives of those that didn't survive.

Example 2:

- **The poster boys/girls of coaching classes.**

These are the students who excel, score high marks, and become the face of success stories.

However, we often overlook the students who didn't achieve similar results.

Survivorship bias makes us believe that following the same path as the poster boys/girls guarantees success.

But remember, there are many students who worked hard but didn't make it into the spotlight.

Takeaway: Be cautious when assuming that the path taken by a few represents the only path to success in coaching classes.

Example 3:

- **Believing in - success = coincidental luck**

Some people attribute success solely to luck, as if it were a random event.

Survivorship bias plays a role here:

We hear stories of successful individuals who credit luck for their achievements.

However, we rarely hear from those who worked hard but didn't achieve the same level of success.

Takeaway: While luck does play a role, it's essential to recognize the effort, resilience, and strategic choices that contribute to success.

Example 4:

- **Dow Jones Industrial Average.**

The Dow is a stock market index that tracks the performance of 30 large U.S. companies.

Over its long history, the Dow has evolved significantly. Companies have been added, removed, and replaced.

Survivorship bias comes into play when we focus only on the current constituents of the Dow without considering the companies that were once part of it but no longer exist.

For example:

The original 12 companies in the Dow (back in 1896) included names like American Cotton Oil and National Lead. Some of these companies faced bankruptcy or were acquired by others.

By ignoring the failed companies, we get a skewed view of the Dow's historical performance.

Takeaway: When analyzing the Dow or any other index, be aware of survivorship bias and consider the full historical context

Example 5:

- **The famous actors and actresses as heroes and award winner.**

A famous example of survivorship bias involves Academy Award-winning actors and actresses.

A study suggested that these award winners lived almost four years longer than their less successful peers.

However, this conclusion overlooks the actors and actresses who didn't win awards or achieve fame.

Remember: The longevity of award winners doesn't necessarily apply to the entire profession

Example 6:

- **Making you the member of the winning team.**

In the context of teams, this bias can lead us to overly optimistic beliefs because we tend to analyze only successful teams.

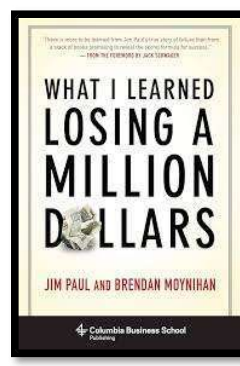
For instance, when interviewing coaches after a championship victory, they often attribute the success to the team's culture.

However, survivorship bias can distort our perception, as we overlook the teams that didn't win but had equally strong cultures.

Takeaway: Winning doesn't always guarantee a good culture; it's essential to consider the full spectrum of teams.

Example 7:

- **Career advice from successful people rather than Unsuccessful**



We often seek advice from successful individuals—CEOs, entrepreneurs, artists, etc.

However, survivorship bias plays a role:

We hear success stories and advice from those who made it to the top.

But what about those who followed similar advice and didn't achieve the same success?

Takeaway: Consider both successful and unsuccessful experiences when shaping your career path. Learn from failures as well as triumphs.

Example 8:

- **The History of Western civilization.**

When studying historical events, we often emphasize the successful civilizations, leaders, and cultural achievements.

But what about the civilizations that declined, the leaders who failed, and the lost cultural treasures?

Survivorship bias can distort our understanding of the past by focusing only on what survived.

Takeaway: To understand history fully, we must consider both successes and failures, recognizing that survivorship bias can lead to incomplete narratives.

THE WAY FORWARD:

1. **Diversify Data Sources:** Don't rely solely on successful examples; include instances of failure as well.
2. **Include Failures in Analysis:** Actively consider both successful and unsuccessful outcomes.
3. **Encourage Open Dialogue:** Foster discussions that include both success stories and failures.
4. **Hear more and more stories.**
5. **Meet people from different areas and sectors**
6. **Travel a lot.**
7. **Be humble.**
8. **Learn from success and learn more from the unsuccessful.**

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3. SUNK COST FALLACY

The Concorde Supersonic Jet:

The Concorde was a British-French supersonic passenger airliner that captured imaginations with its sleek design and the promise of ultra-fast travel. It made its first commercial flights in 1976, becoming the first jet to exceed the speed of sound in civil aviation history. However, despite its technological marvel, the Concorde faced significant economic challenges from the outset.

Economic Limitations:

The Concorde project suffered from wrong marketing choices, exorbitant production costs, high maintenance expenses, and steep fuel prices.

Its ticket prices were prohibitively expensive, leading to low demand. The project was financially unsustainable from the beginning.

Sunk Costs and Decision Dilemmas:

Imagine you're the manager of Concorde's R&D division.

You face two scenarios:

Scenario 1: You've already invested €900 million in the project and need an additional €100 million to complete it.

Scenario 2: You haven't invested anything yet and must decide whether to allocate €100 million.

Intuitively, you might be tempted to say yes in the first case and no in the second. However, both situations are equivalent in terms of profitability.

The Sunk Cost Fallacy (Concorde Fallacy):

The French and British governments, having heavily invested money and prestige in the Concorde project, continued to pour funds into it—even when it became clear that it wouldn't be financially viable.

This cognitive bias, known as the sunk cost fallacy, occurs when people justify increased investment based on cumulative prior investment.

Essentially, they consider non-recoverable costs (sunk costs) as a rationale for continuing, even when it's irrational.

Opportunity Cost Ignored:

The sunk cost fallacy blinds decision-makers to the opportunity cost—the value of the next best alternative foregone.

By persisting with the Concorde despite its lack of economic viability, the governments missed opportunities to invest in more promising ventures.

Lessons Learned:

The Concorde fallacy reminds us to look beyond sunk costs and consider the overall picture. Sometimes, cutting losses and redirecting resources toward better prospects is the wiser choice.

WHAT IS SUNK COST FALLACY?

- Sunk cost fallacy leads people to make irrational decisions based on past investments (sunk costs) rather than objectively evaluating the current situation.

- Individuals continue investing time, money, or effort into a project, relationship, or endeavor simply because they've already committed resources to it.

- Instead of considering future benefits and costs.

Example 1:

- **Non-Refundable ticket to movie or concert.**

You're continuing to invest time in something that isn't providing value simply because you've already spent money on it.

Example 2:

- **Any course you started for long time.**

Important is what are you learning which will impact what things of your life.

Example 3:

- **Relationship / friendship.**

The friendship has become toxic. Perhaps you've grown apart, or there are unresolved conflicts. Despite the challenges, you hesitate to let go because of all the history you've built together. This is the sunk cost fallacy at play. You're reluctant to walk away because you fear that all those years would be wasted if you end the friendship now.

Example 4:

- **The collection of any specific items – Eg. Cards, mobile phones, collectibles.**

Maybe the thrill of the hunt has faded, or you're running out of storage space. Despite this, you hesitate to stop collecting because of all the resources you've poured in. The sunk cost fallacy whispers, "You can't quit now; think of all those rare cards you've acquired!" So, you keep buying booster packs, even when it no longer brings joy.

THE WAY FORWARD

- **Beware of 'ENDOWMENT EFFECT'** – for example –

  Imagine you've owned a car for a while. During that time, you've become emotionally attached to it, and this attachment influences your perception of its value. When you decide to sell the car, you might ask for a higher price than what you'd be willing to pay if you were the buyer. Essentially, you believe your car is unique and worth more, even if it's a common model.

  Car dealerships sometimes leverage the endowment effect by offering free trials. Prospective buyers can take a car home, drive it for a few days, and experience ownership. This temporary sense of ownership often leads buyers to value the car more, potentially increasing the likelihood of a sale.

  So, next time you find it hard to part with your old car, remember that your emotional attachment might be affecting your perception of its worth!

- **Recognize Sunk Costs:** Understand that past investments are unrecoverable. Don't let them dictate future decisions.

- **Reframe Perspective:** Focus on present and future benefits rather than dwelling on past losses.

- **Consider Opportunity Cost:** Evaluate alternative uses for your resources. What else could you do with the time, money, or effort?

- **Set Clear Criteria:** Establish decision-making criteria independent of sunk costs. Stick to these guidelines.

- **Emotional Detachment:** Separate emotions from the decision. Be objective and pragmatic.

- **Learn from Mistakes:** Use past experiences to improve future choices. Don't let sunk costs cloud your judgment.

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4. Hindsight Bias

Definition:
Hindsight bias, often called the *"I-knew-it-all-along" effect*, is a psychological tendency where people believe—after an event has occurred—that they had predicted or expected it all along. In simple terms, it's our mind's way of rewriting history to make the past look more predictable than it actually was.

It's a common human bias that makes us feel smarter than we were *before* the event happened. We forget how uncertain things truly were in the moment.

Explanation with Examples:

1. **The Cuban Missile Crisis (1962):**
   After the U.S. and the Soviet Union narrowly avoided nuclear war, many people later claimed that the peaceful resolution was obvious—that both sides would "of course" back down. But was it really that clear at the time?
   Not at all. The world was on the edge of nuclear confrontation; leaders on both sides faced immense uncertainty and fear. Yet, when we look back *knowing the outcome*, our brain conveniently reconstructs the story as if peace was always the inevitable ending.

This is hindsight bias at work—it gives us the false comfort that history was bound to happen the way it did.

2. **The 9/11 Attacks (2001):**
   After 9/11, many said the attacks "could have been predicted" because of earlier intelligence warnings or failed signals. But in truth, before the event, those pieces of information were scattered, ambiguous, and buried among thousands of other threats.
   Only *after* the event, when the dots are already connected, do we think the pattern was obvious. That's the illusion created by hindsight bias—it makes complex events look simple *after* they've unfolded.

Why It Matters (for Students):
Hindsight bias affects not just historians or analysts but also students, researchers, and policymakers. It shapes how we judge others' decisions, how we interpret data, and how we learn from mistakes. For example:

- In politics, it may lead to unfairly blaming leaders for "not foreseeing" crises.
- In research, it can make us overconfident about our theories after seeing the results.
- In personal life, we might think, "I knew I shouldn't have trusted that person," though we didn't actually know it at the time.

Way Forward:

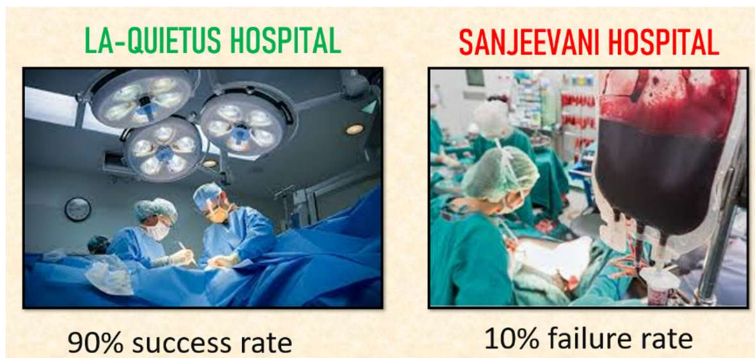
1. **Acknowledge uncertainty:**
   Before judging decisions of the past, remind yourself—people didn't have the same information then that we have now.
2. **Keep a record of predictions:**
   Writing down what you *actually* think before outcomes happen helps you see later how hindsight changes perception.
3. **Analyze process, not just outcome:**
   Ask: "Was the decision rational given what was known *then*?" instead of "Did it turn out right?"
4. **Train critical thinking:**
   Students should consciously separate the *event as it happened* from the *story we tell about it afterward.*

In short:
Hindsight bias is our brain's historian—keen to make the messy, uncertain past look like a neat and predictable story. But real learning begins when we stop saying, *"I knew it all along,"* and start asking, *"Why did it seem so uncertain back then?"*

5. Framing effect Bias

Definition:
The *framing effect* is a cognitive bias where people's decisions and judgments change depending on *how information is presented*, rather than what the information actually is. In other words, *the way a choice is framed*—as a gain or a loss, positive or negative—can dramatically alter our response, even when the underlying facts remain the same.

Our brains are not purely logical calculators; they are emotional storytellers. We respond not to *data*, but to *how the data feels*.



Explanation with Examples:

1. **The Medical Treatment Example:**
   Imagine a doctor tells you:

- *Option A:* "This treatment has a 90% survival rate."
- *Option B:* "This treatment has a 10% death rate."

Both statements convey the exact same fact. But most people feel much safer with Option A—because the *frame* highlights survival, not death. That's the framing effect at work: our reaction depends on whether the outcome is framed in positive or negative language.

2. **The COVID-19 Pandemic Example:**
   During the pandemic, governments often had to frame public health messages carefully. Saying *"If we don't act, 1 lakh people could die"* sounds far more alarming than *"If we act, we can save 99% of our population."*
   Both might be mathematically true—but the emotional pull is very different. Policymakers understood that the *frame* could influence public behavior more than the facts themselves.

3. **The War and Peace Example – The Gulf War (1991):**
   When leaders presented military intervention as a way to *"liberate Kuwait"* (positive frame), public support was high. Had it been framed as *"sending young soldiers to die in a distant desert"* (negative frame), public opinion might have been very different.
   The moral? Humans are persuaded not by *logic alone*, but by *how logic is dressed*.

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Why It Matters (for Students):
The framing effect teaches us that **language is power**. The same issue—be it war, economy, or climate change—can be made to sound heroic or horrifying depending on the chosen frame.

For instance:

- In economics: “Unemployment is down to 5%” vs. “1 in 20 people can’t find work.”
- In social studies: “Reservation empowers marginalized groups” vs. “Reservation limits opportunities for others.”

Both are true, yet each shapes a completely different emotion and argument. Students of politics, sociology, and media must constantly ask—*“What frame is being used here?”*

Way Forward:

1. **Be conscious of language:**
   Before reacting to data or news, reframe it in your own words. Ask, *“How would this sound if it were phrased the opposite way?”*
2. **Use neutral framing in analysis:**
   When studying political debates or media coverage, separate *facts* from *frames*. Recognize emotional manipulation disguised as information.
3. **Experiment with framing:**
   As a thinker or writer, try framing the same statement in multiple ways. You’ll see how easily tone and wording can shift perception.
4. **Cultivate “Frame-awareness”:**
   In every argument, before choosing sides, pause and ask—*“Am I reacting to the fact or to the frame?”*

In short:
The *framing effect* is like holding truth in two mirrors—each reflection looks different, but the object is the same. Once we recognize this bias, we stop being passive consumers of information and start becoming active interpreters of meaning.

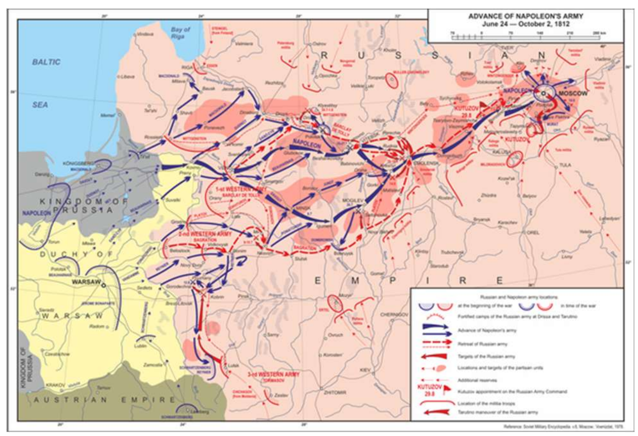
6. Action Bias

Definition:
Action bias is the human tendency to prefer **doing something** over **doing nothing**, even when inaction might actually be the wiser or safer choice. It stems from the discomfort we feel in waiting, watching, or accepting uncertainty. We equate action with control and courage—forgetting that sometimes, stillness is the highest form of strategy.

In short: ***when faced with uncertainty, our instinct is to act, not to reflect.***

Explanation with Example:

1. **Napoleon's Attack on Russia:**
   Napoleon Bonaparte, one of history's most brilliant military minds, fell victim to the very bias that topples great leaders—the *action bias.*



After conquering much of Europe, Napoleon turned his eyes toward Russia. His advisors warned him about the risks: the vast distances, the brutal winters, the long supply lines, and the unpredictability of Russian strategy. But Napoleon couldn't stay still. He believed that *inaction would mean weakness*.

And so, he *acted*—launching one of the largest invasions in history with over 600,000 troops. The result? Disaster. The Russian "scorched-earth" strategy, harsh winter, and stretched logistics wiped out almost his entire army.

In hindsight, patience, diplomacy, or limited engagement might have preserved his empire. But Napoleon's belief that "decisive action is always strength" blinded him to the wisdom of waiting. His downfall began not with defeat, but with **impulsive action.**

This is action bias: the compulsion to *do something*, even when doing nothing is the better move.

Other Simple Illustrations:

- In sports, goalkeepers often dive left or right during penalty kicks, though statistically, staying in the center might save more goals. But standing still looks foolish—so they move.
- In politics, leaders announce hasty policies during crises just to *appear decisive*, even if it worsens the problem.
- In daily life, students rush to respond in exams or debates, fearing silence might seem like ignorance—when sometimes, a pause and reflection would lead to a better answer.

Why It Matters (for Students):
Action bias is dangerous because it hides under the mask of *confidence*. Society praises "doers," not "thinkers," and that cultural bias makes us overlook the power of strategic restraint.
Students, researchers, and policymakers must learn that *inaction is not laziness*—it is sometimes clarity.

Napoleon's Russia campaign is the ultimate classroom case: a great mind, undone not by ignorance, but by impatience.

Way Forward:

1. **Pause before action:**
   Ask: *"Am I acting to solve the problem—or just to feel in control?"*
2. **Learn the art of strategic stillness:**
   True leadership sometimes lies in *non-action*—the wisdom of waiting until conditions are right.
3. **Reframe inaction as decision:**
   Doing nothing is not "failure to decide." It *is* a decision—just one rooted in patience and prudence.
4. **Reflect on feedback loops:**
   Study past decisions where impulsive action led to poor outcomes. History, from Napoleon to corporate crises, is full of them.

In short:
Action bias seduces us with the illusion of control. Napoleon marched to Moscow not because he had to—but because he couldn't bear to *not* march. The result? He lost what he was trying to secure through action—his empire.

So remember: **Sometimes the wisest move is not the next step forward, but the conscious step back.**